# CHENYANG LAI

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Research Fellow, Department of Mechanical and Aerospace Engineering
Nanyang Technological University

## RESEARCH INTERESTS

- Physics-informed Machine Learning (PIML)
- Industrial AI
- AI Safety and Trustworthiness
- Manufacturing Process Modeling and Monitoring
- Prognostics and Health Management (PHM)

## EMPLOYMENT

**Research Fellow** 05/2026 – Present
Nanyang Technological University (NTU), Singapore
Mentor: Dr. Xuan Liang

- Thermal and deformation-field prediction for advanced additive manufacturing using PIML methods.

**Postdoctoral Researcher** 02/2025 – 05/2026
Politecnico di Milano (POLIMI), Milan, Italy
Mentors: Prof. Piero Baraldi and Prof. Enrico Zio

- Condition monitoring and degradation assessment of power electronics used in wind turbines.
- Anomaly detection of power transistors using scattering parameters.
- Reinforcement learning for limp-mode management of power converters.

## EDUCATION

**Politecnico di Milano (POLIMI)** Milan, Italy
Ph.D. in Energy and Nuclear Science and Technology, *Cum Laude* 01/2021 – 07/2025
Supervisors: Prof. Piero Baraldi and Prof. Enrico Zio

- Thesis: Physics-informed deep learning methods of prognostics and health management for predictive maintenance of industrial equipment

**Northwestern Polytechnical University (NPU)** Xi'an, China
Master of Control Science and Engineering, *Cum Laude* 08/2017 – 04/2020

**Harbin Engineering University (HEU)** Harbin, China
Bachelor of Science in Energy and Power Engineering 08/2013 – 07/2017

## RESEARCH PROJECTS

**Ministry of Education (MOE) Project, Singapore** 05/2026 – Present
*Physics-Enhanced Machine Learning-based Simulation with Applications to Structural Design and Process Optimization for Metal Additive Manufacturing*

- Transfer-learning-enhanced physics-informed neural network for unsupervised full-field distortion prediction of metal additive manufacturing by directed energy deposition.
- Physics-informed neural network for thermal-field prediction of the directed energy deposition process.

**European Union Project: SAFEPOWER** 01/2025 – 06/2026
*Safer and More Reliable WBG/UWBG-Based MVDC Power Converters*

- AI methods for degradation assessment and remaining useful life prediction of IGBTs in wind turbine converters, coordinating with the University of Warwick, UK.
- Convolutional neural network to predict the degradation evolution of IGBT devices from S-parameters, coordinating with the University of Technology Tarbes Occitanie Pyrenees (UTTOP), France.
- Reinforcement learning framework for limp-mode management of power converters, coordinating with Power Electronics España, S.L.

**European Union Project: MOSAIC** 04/2025 – 06/2026
*A Mosaic of Essential Electronic Components and Systems for Our Automated Digital Future in Industry and Mobility*

- Zero-shot anomaly detection for semiconductor manufacturing based on contrastive language–image pre-training (CLIP), coordinating with Infineon Technologies Austria AG.

**European Union Project: iRel40** 01/2021 – 04/2023
*Intelligent Reliability 4.0*

- Feature selection and AI methods for case-temperature estimation of IGBTs used in induction cooktops, coordinating with BSH Electrodomésticos España, Spain.
- Physics-informed domain adaptation and AI methods for health assessment and remaining useful life prediction of electronic components, coordinating with BSH Electrodomésticos España, Spain.

## JOURNAL PUBLICATIONS

12. **Lai, C.**, & Liang, X.*. (2026). Transfer learning-enhanced PINN for unsupervised deformation prediction of directed energy deposition. In preparation.
11. **Lai, C.**, Baraldi, P.*, & Zio, E. (2026). PIDEL: Physics-Informed Deep Evidential Long Short-Term Memory Networks for Remaining Useful Life Prediction with Uncertainty Quantification of Electronic Components. In preparation.
10. **Lai, C.***, Liang, X., Ran, L., & Zio, E. (2026). Few-Shot Anomaly Detection under Data Scarcity Using Pretrained Vision-Language Representations and Dropout-Calibrated Decision Boundaries. *Advanced Engineering Informatics*. Under review.
9. **Lai, C.**, Baraldi, P.*, Baldo, L., Bertone, M., Dalla Vedova, M. D. L., Quattrocchi, G., & Zio, E. (2026). Physics-informed long short-term memory networks with transfer learning for anomaly detection in electromechanical actuators. *Engineering Applications of Artificial Intelligence*. Under review.
8. Liu, A., Qu, C., **Lai, C.**, Zhu, H., & Wang, B.* (2026). SE-PLNN: A sample entropy and parallel liquid neural network framework for Remaining Useful Life prediction of mechanical systems under multi-fault degradation. *Measurement*, 292, 123459. (IF = 6.1)
7. Li, X., Lin, W.*, Wang, B., Shi, L., **Lai, C.**, Qiao, Q., et al. (2026). Comprehensive compound fault modelling and characterisation of building integrated photovoltaics based on thermal-electrical coupled simulation. *Solar Energy*, 315, 114790. (IF = 7.9)
6. **Lai, C.***, Baraldi, P.*, & Zio, E. (2026). Physics-informed domain adaptation for health state assessment of industrial components. *Reliability Engineering & System Safety*, 113021. (IF = 13.7)
5. Liu, H., Li, B.*, **Lai, C.**, & Zio, E. (2026). Unsupervised domain adaptation regression network with efficient channel attention mechanism for tool wear prediction under variable cutting conditions. *Journal of Manufacturing Processes*, 172, 1091–1105. (IF = 7.8)
4. **Lai, C.***, Wang, B.*, Fan, S., & Zio, E. (2026). Self-balancing physics-informed LSTM for soft-sensing of slurry concentration in dredger operation. *Advanced Engineering Informatics*, 69, 103978. (IF = 11.5)
3. **Lai, C.**, Baraldi, P.*, & Zio, E. (2025). Gradient-enhanced physics-informed long short-term memory networks for stable and accurate prediction of the RUL of electronic components. *Reliability Engineering & System Safety*, 111485. (IF = 13.7)
2. **Lai, C.**, Baraldi, P.*, & Zio, E. (2024). Physics-informed deep autoencoder for fault detection in new-design systems. *Mechanical Systems and Signal Processing*, 215, 111420. (IF = 10.2)
1. **Lai, C.**, Ahmed, I.*, Zio, E.*, Li, W., Zhang, Y., Yao, W., & Chen, J. (2024). A multistage physics-informed neural network for fault detection in regulating valves of nuclear power plants. *Energies*, 17(11), 2647. (IF = 3.9)

## CONFERENCE AND WORKSHOPS

15. **Lai, C.**, Baraldi, P.*, Moyo, W., Baffreau, S., Liang, X.*, Vidal, P., Vine, G., & Zio, E. (2026). Unsupervised Degradation Evolution Tracking of Power Transistors Using S-Parameters based on Convolutional Neural Network Autoencoders. *Procedia CIRP* (CIRPe 2026, Nanyang Technological University, Singapore). (Proceeding & Presentation)
14. Liu, H., Shen, M., **Lai, C.***, & Liu, Z. (2026). PIASANet: Physics-informed adaptive spectral attention network for cross-load bearing fault diagnosis. *Proceedings of the 10th International Conference on Reliability Engineering (ICRE)*. (Proceeding & Presentation)
13. **Lai, C.**, Baraldi, P.*, Zinchenko, O., Zhang, X., Ran, L., & Zio, E. (2026). Assessment of the degradation of IGBTs used in wind turbine power converters by a method based on physics-informed transformers. *Proceedings of the 36th European Safety and Reliability Conference (ESREL 2026)*, 3113–3120. (Proceeding & Presentation)
12. **Lai, C.**, Baraldi, P.*, Aruna, Baldo, L., Dalla Vedova, M. D. L., Quattrocchi, G., & Zio, E. (2025). Anomaly detection in power inverters of electromechanical actuators based on convolutional neural network and long short-term memory cells. *2025 9th International Conference on System Reliability and Safety (ICSRS)*, IEEE, 136–142. (Proceeding & Presentation)
11. **Lai, C.***, Baraldi, P., & Zio, E. (2025). AI strategies in C&HM for MVDC converters: Studies ongoing. *SAFEPOWER WP3 Technical Workshop*. (Presentation)
10. **Lai, C.**, Baraldi, P.*, Quattrocchi, G., Dalla Vedova, M. D. L., Baldo, L., Bertone, M., & Zio, E. (2024). Detection of abnormal conditions in electro-mechanical actuators by physics-informed long short-term memory networks. *PHM Society European Conference*, 8(1). (Proceeding & Presentation)
9. **Lai, C.**, Baraldi, P.*, Quattrocchi, G., Dalla Vedova, M. D. L., Baldo, L., Bertone, M., & Zio, E. (2023). Fault Detection in Electro-Mechanical Actuators by Physics-Informed Long Short-Term Memory Networks. *2023 7th International Conference on System Reliability and Safety (ICSRS)*, IEEE. (Abstract & Presentation)
8. **Lai, C.***, Baraldi, P., Ahmed, I.*, & Zio, E. (2022). Development of PHM Method for Power Electronic Devices Based on Physics-Informed Neural Networks. *iRel40 Use Case Workshop, Ljubljana, Slovenia.* (Poster & Presentation)
7. **Lai, C.**, Baraldi, P.*, Floreale, G., & Zio, E. (2022). Physics-Informed Long Short-Term Memory Network for the Prediction of the Remaining Useful Life (RUL) of Insulated-Gate Bipolar Transistors (IGBTs). *2022 6th International Conference on System Reliability and Safety (ICSRS)*, IEEE. (Abstract & Presentation)
6. Juybari, M. N., Ahmed, I.*, Baraldi, P., Del Cueto, A., Gil, J., **Lai, C.**, et al. (2022). Estimation of the case temperature of insulated gate bipolar transistors in induction cooktops by deep neural network. *2022 6th International Conference on System Reliability and Safety (ICSRS)*, IEEE, 259–263. (Proceeding & Presentation)
5. **Lai, C.**, Baraldi, P.*, Ahmed, I., Zio, E., Gil, J., & Llorente, S. (2022). Monitoring degradation of insulated gate bipolar transistors in induction cooktops by artificial neural networks. *Proceedings of the 32nd European Safety and Reliability Conference (ESREL 2022)*, 1338–1345. (Proceeding & Presentation)
4. **Lai, C.**, Baraldi, P.*, & Zio, E. (2022). A Physics-informed Autoencoder for Early Fault Detection of Electro-hydraulic Actuators. *2021 5th International Conference on System Reliability and Safety (ICSRS)*, IEEE. (Abstract & Presentation)
3. Mao, H., Guo, Y., Li, R., & **Lai, C.** (2019). Versatile simulation platform for turboshaft engine control system. *2019 Chinese Control Conference (CCC)*, IEEE, 7211–7216. (Proceeding & Presentation)
2. **Lai, C.**, Guo, Y., & Mao, H. (2019). Main fuel pump health monitoring method based on Kriging surrogate model. *2019 Chinese Control Conference (CCC)*, IEEE, 5112–5117. (Proceeding & Presentation)
1. Zhao, W., Guo, Y., & **Lai, C.** (2018). Sensor fault diagnosis and estimation based on multiple-model approach for aeroengine. *2018 IEEE CSAA Guidance, Navigation and Control Conference (CGNCC)*, 1–6. (Proceeding & Presentation)

## AWARDS

- Best Presentation Award, Anomaly detection in power inverters of electromechanical actuators based on convolutional neural network and long short-term memory cells, 9th International Conference on System Reliability and Safety (ICSRS), IEEE, 2025.
- Best Presentation Award, Fault detection in electro-mechanical actuators by physics-informed long short-term memory networks, 7th International Conference on System Reliability and Safety (ICSRS), IEEE, 2023.
- Outstanding Postgraduate Award, Northwestern Polytechnical University, 2020.
- Second Prize in the National Postgraduate Mathematical Modeling Competition, 2018.

## ACADEMIC ACTIVITIES

- Chair, session "T1-C: Energy, Sustainability & Remanufacturing", CIRPe 2026, Nanyang Technological University, Singapore, October 13–16, 2026.
- Chair, session "T2-B: Circular Manufacturing, Remanufacturing & Handling Systems", CIRPe 2026, Nanyang Technological University, Singapore, October 13–16, 2026.
- Invited speaker, Symposium on "AI-Driven Simulation, Design and Control for Manufacturing Processes", 2nd International Conference on Intelligent Manufacturing (ICIM), Singapore, July 27–31, 2026.
- Chair, special session "Physics-Informed Machine Learning for Prognostics and Health Management of Industrial Equipment", 10th International Conference on Reliability Engineering (ICRE), Hangzhou, China, July 19–21, 2026.
- Co-chair, 10th International Conference on Reliability Engineering (ICRE), Hangzhou, China, July 19–21, 2026.
- Chair, special session "SS-06: Physics-Informed Machine Learning for Reliability, Availability, Maintainability and Safety (RAMS)", European Safety and Reliability Conference (ESREL), Braga, Portugal, June 14–19, 2026.
- Chair, special session "SS-10: Development and Application of Methods for Enhancing the Reliability of Power Electronics Devices", European Safety and Reliability Conference (ESREL), Braga, Portugal, June 14–19, 2026.
- Invited scholar, Silk Road International Symposium (SRIS), Xi'an Jiaotong University, Xi'an, China, November 1–4, 2025.
- Journal Reviewer for:
  *Mechanical Systems and Signal Processing*
  *Reliability Engineering & System Safety*
  *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*
  *IEEE Transactions on Industrial Informatics*
  *IEEE Transactions on Transportation Electrification*
  *IEEE Journal of Emerging and Selected Topics in Power Electronics*
  *Applied Soft Computing*

## TEACHING AND MENTORING

- Lecturer, continuing education course "RAM&PHM 4.0: Advanced Methods for Reliability, Availability, Maintainability, Prognostics and Health Management of Industrial Equipment", Politecnico di Milano, 2022, 2023, and 2025.
- Lecturer, "Artificial Intelligence and Advanced Simulation for the Safety, Reliability and Maintenance of Energy Systems", M.Sc. in Nuclear Engineering, Politecnico di Milano, 2022, 2023, and 2024.

### Mentoring and supervision

- 2026–Present: Francesco Marco Riboli (M.Sc.) — AI-based degradation assessment for power electronic devices using scattering parameters.
- 2026–Present: Yunlu Bai (Ph.D.) — Physics-informed machine learning for high-altitude structural health monitoring.
- 2025–2026: Tibyan Gadalla Naeem Mustafa (M.Sc.) — Physics-informed neural networks for modelling high-burnup structure formation in nuclear fuel.
- 2023–2024: Aruna (M.Sc.) — Fault detection of power inverters in electromechanical actuators using a CNN-LSTM approach.
- 2023–2024: Camilla Battistini (M.Sc.) — Physics-informed neural networks for the safety analysis of nuclear reactors.

## SKILLS

- **Languages:** Mandarin (native), English (working proficiency), Italian (limited working proficiency), Spanish (elementary).
- **Computing:** MATLAB/Simulink, Python, PyTorch, C/C++, Git, Amesim, Ansys, LTspice, LaTeX, Microsoft Office.
- **Soft skills:** self-learning, critical thinking, teamwork, problem solving.